

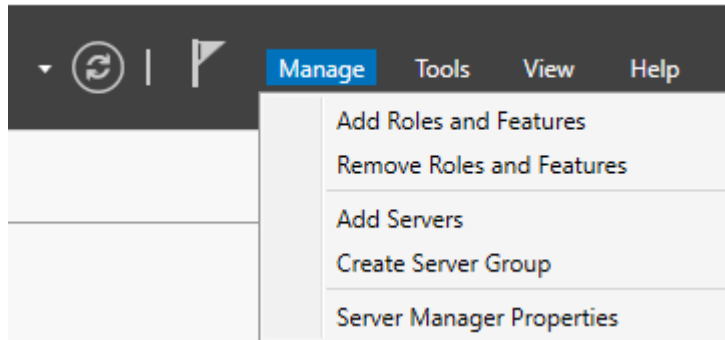
Windows Lab 2

By Jai Pandey (260840140011)

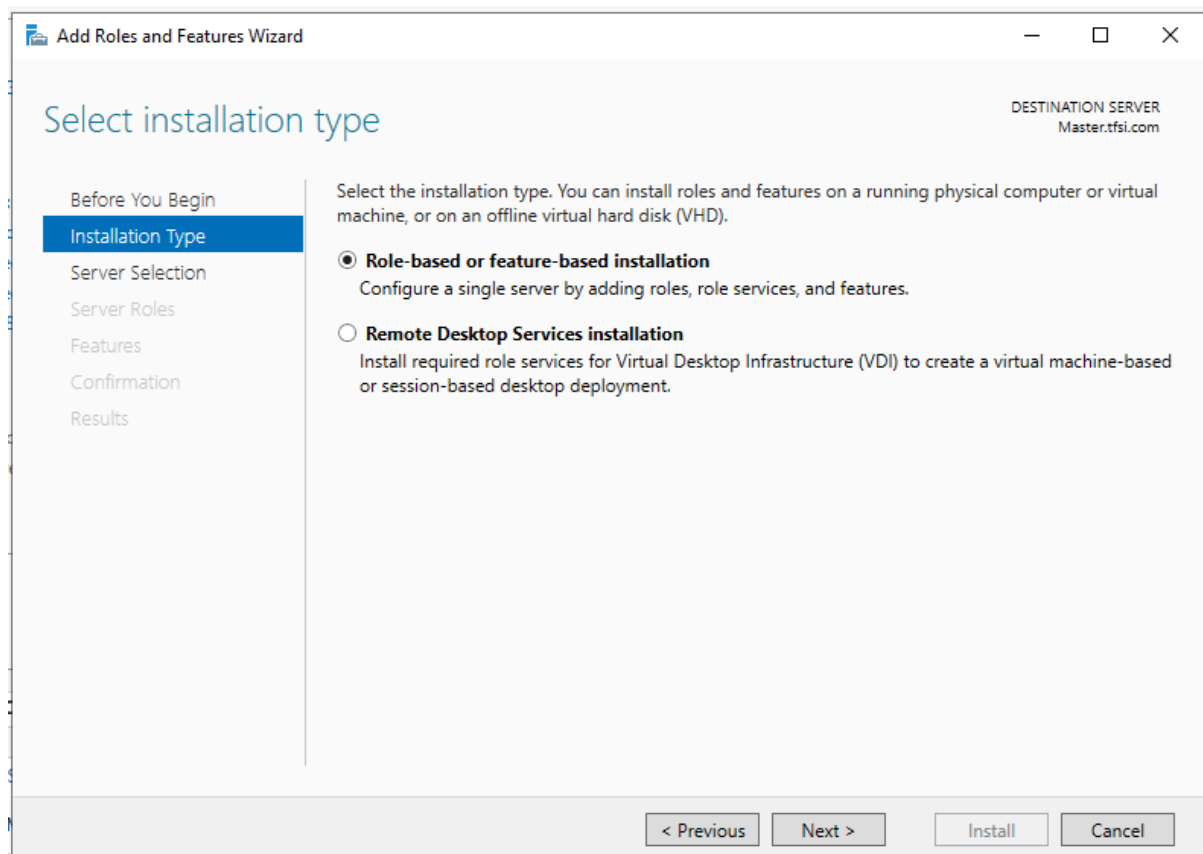
Question 1

1. Adding the DNS Feature

Server Manager > Manage > Add Roles and Features



Select Role Based or feature based installation



Select destination server

Add Roles and Features Wizard

DESTINATION SERVER
Master.tfsi.com

Select destination server

Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results

Select a server or a virtual hard disk on which to install roles and features.

☒ Select a server from the server pool
☐ Select a virtual hard disk

Server Pool

Filter:

Name	IP Address	Operating System
Master.tfsi.com	192.168.200.100	Microsoft Windows Server 2022 Datacenter Evaluation

1 Computer(s) found

This page shows servers that are running Windows Server 2012 or a newer release of Windows Server, and that have been added by using the Add Servers command in Server Manager. Offline servers and newly-added servers from which data collection is still incomplete are not shown.

< Previous Next > Install Cancel

Select DNS Server

Add Roles and Features Wizard

DESTINATION SERVER
Master.tfsi.com

Select server roles

Before You Begin
Installation Type
Server Selection
Server Roles
Features
Confirmation
Results

Select one or more roles to install on the selected server.

Roles

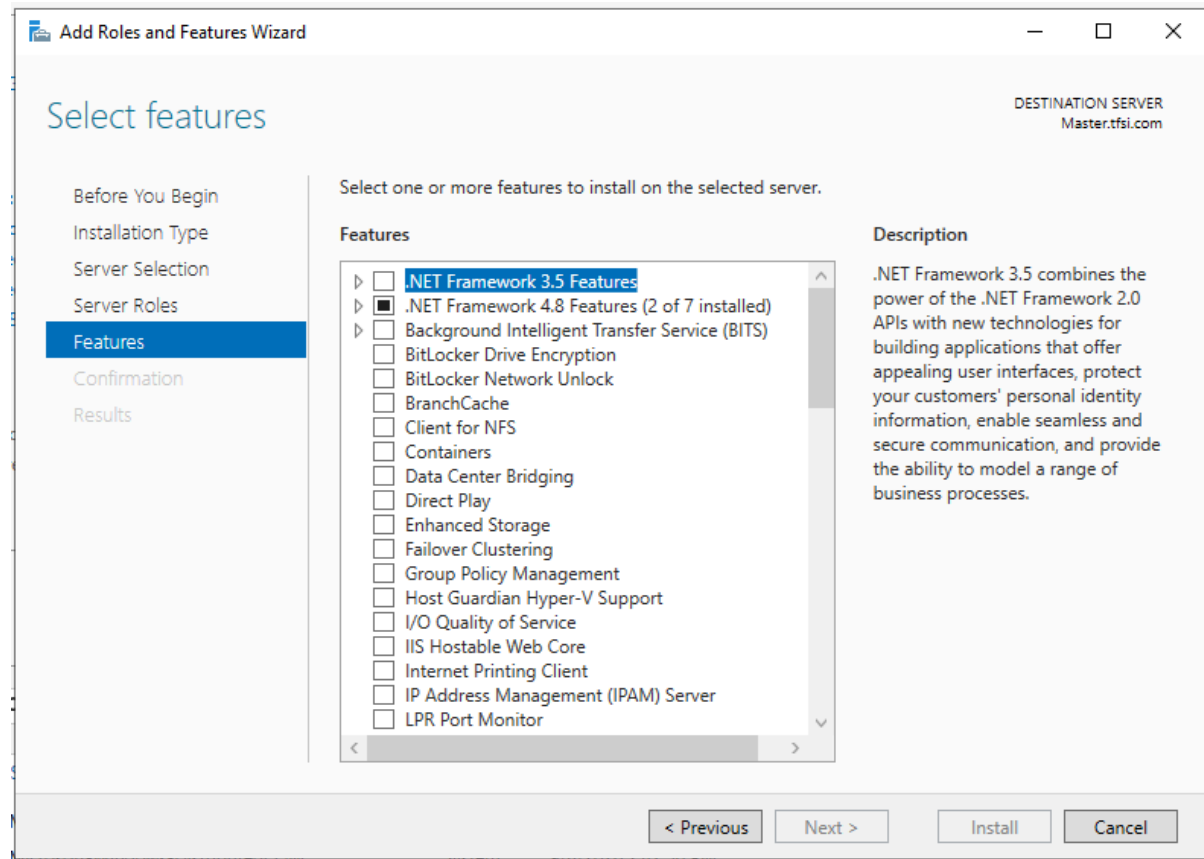
- ☐ Active Directory Certificate Services
- ☐ Active Directory Domain Services
- ☐ Active Directory Federation Services
- ☐ Active Directory Lightweight Directory Services
- ☐ Active Directory Rights Management Services
- ☐ Device Health Attestation
- ☐ DHCP Server
- ☒ **DNS Server (Installed)**
- ☐ Fax Server
- ☒ File and Storage Services (1 of 12 installed)
- ☐ Host Guardian Service
- ☐ Hyper-V
- ☐ Network Controller
- ☐ Network Policy and Access Services
- ☐ Print and Document Services
- ☐ Remote Access
- ☐ Remote Desktop Services
- ☐ Volume Activation Services
- ☐ Web Server (IIS)
- ☐ Windows Deployment Services

Description

Domain Name System (DNS) Server provides name resolution for TCP/IP networks. DNS Server is easier to manage when it is installed on the same server as Active Directory Domain Services. If you select the Active Directory Domain Services role, you can install and configure DNS Server and Active Directory Domain Services to work together.

< Previous Next > Install Cancel

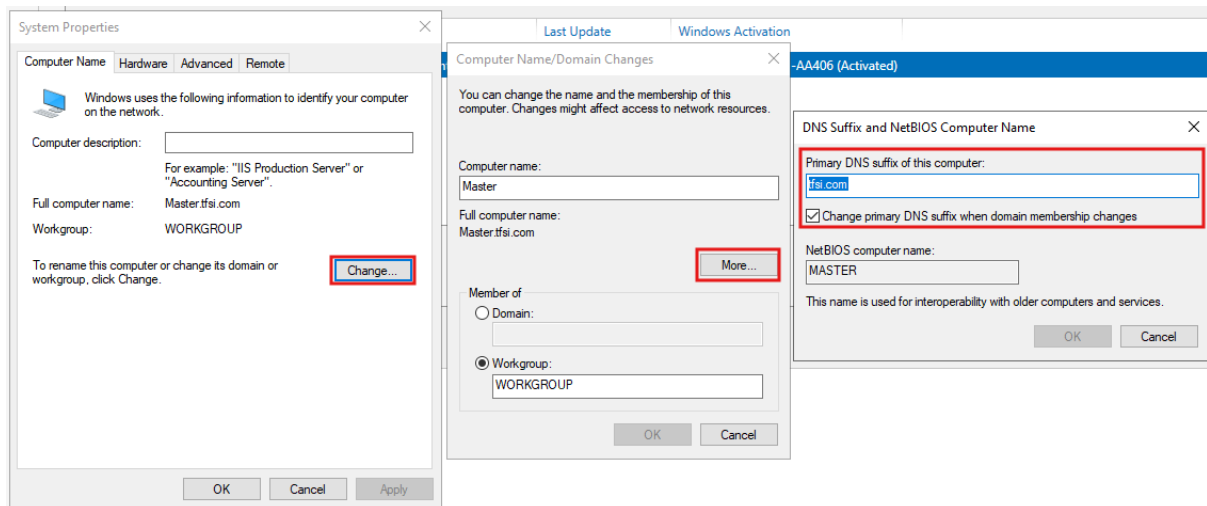
Skip selecting other features .



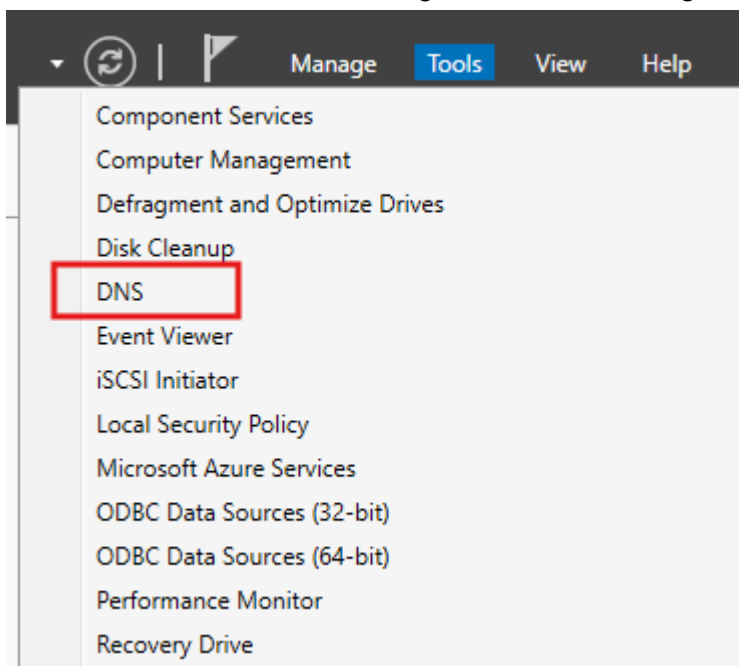
After we restart, we can proceed further.

To configure the DNS server, we need to change our domain

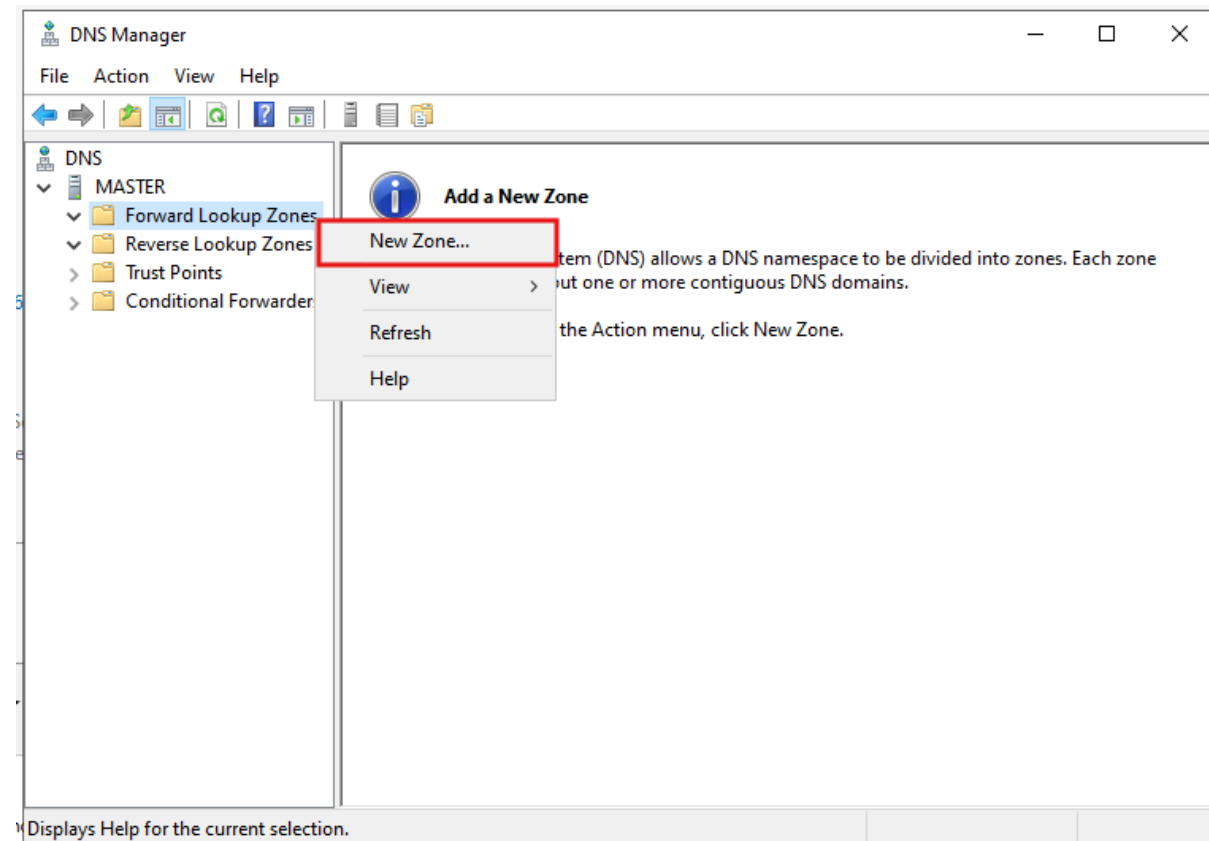
2. Server Manager > Local Server > Computer Name > Change > More > Enter new Primary DNS Suffix.



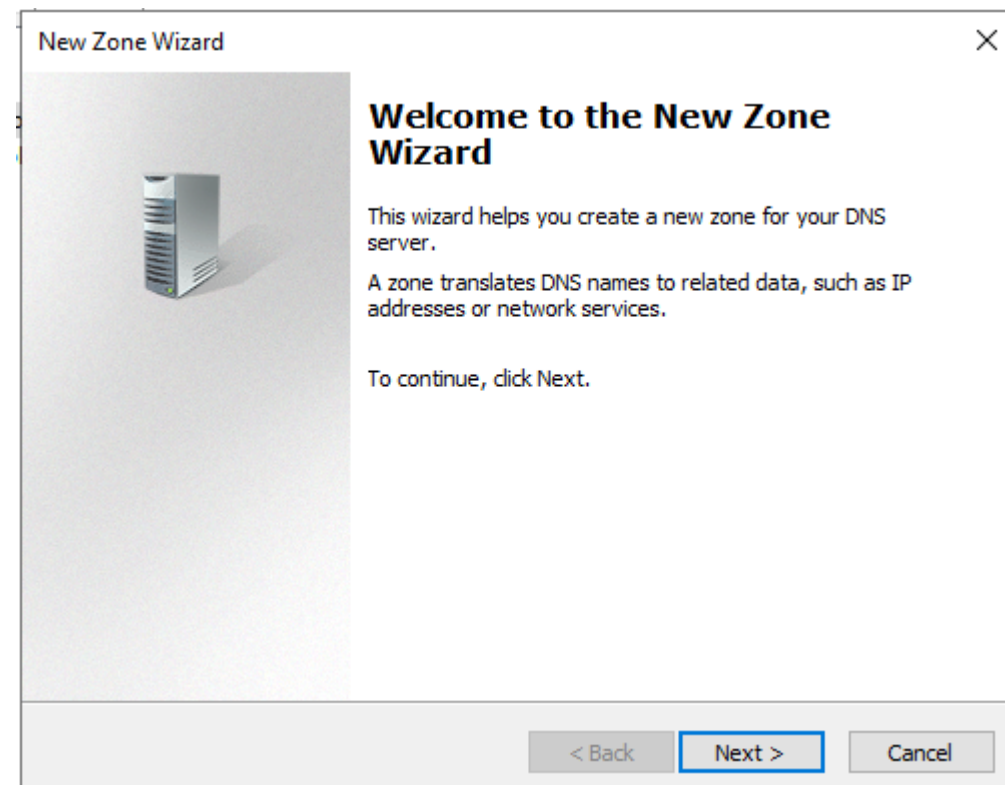
Now we can see the DNS Manager under the 'Manage' option in the top bar.



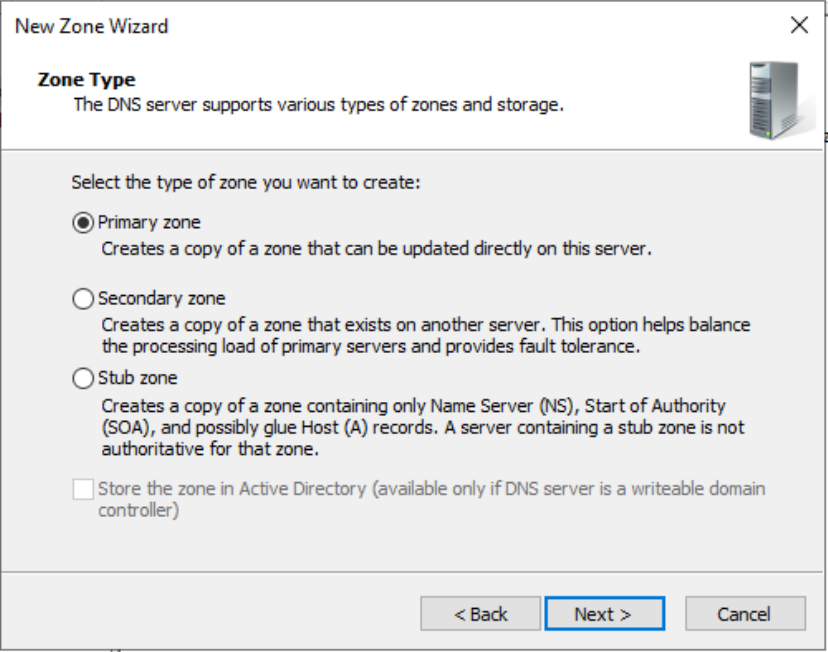
DNS Manager > DNS > Computer Name > Forward Lookup Zones
Right click and click on 'New Zone'



Create a new zone using the new zone wizard



Primary Zone



New Zone Wizard

Zone Type
The DNS server supports various types of zones and storage.

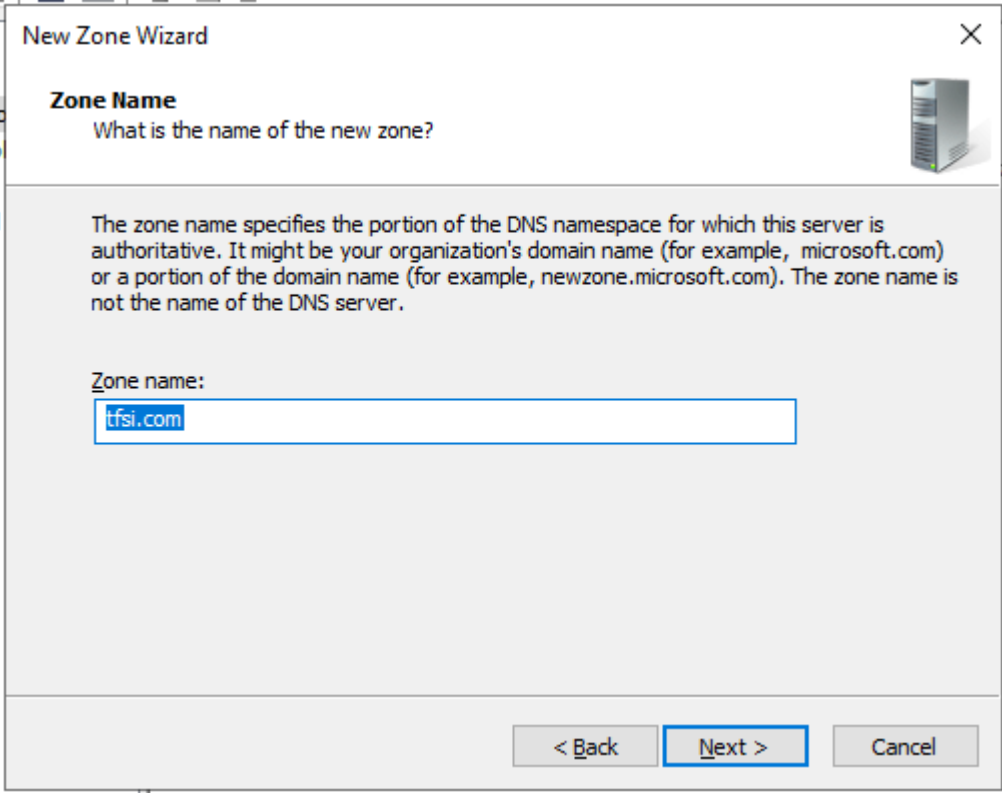
Select the type of zone you want to create:

- ☒ Primary zone
Creates a copy of a zone that can be updated directly on this server.
- ☐ Secondary zone
Creates a copy of a zone that exists on another server. This option helps balance the processing load of primary servers and provides fault tolerance.
- ☐ Stub zone
Creates a copy of a zone containing only Name Server (NS), Start of Authority (SOA), and possibly glue Host (A) records. A server containing a stub zone is not authoritative for that zone.

☐ Store the zone in Active Directory (available only if DNS server is a writeable domain controller)

< Back Next > Cancel

Keep zone name same as Domain



New Zone Wizard

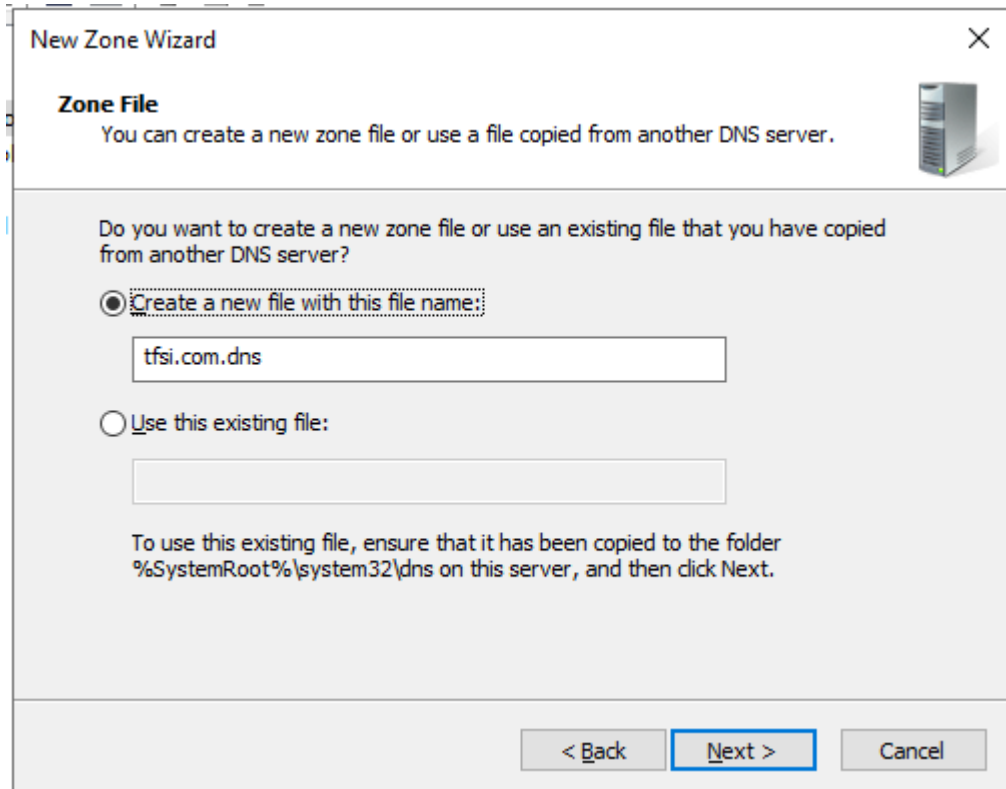
Zone Name
What is the name of the new zone?

The zone name specifies the portion of the DNS namespace for which this server is authoritative. It might be your organization's domain name (for example, microsoft.com) or a portion of the domain name (for example, newzone.microsoft.com). The zone name is not the name of the DNS server.

Zone name:

< Back Next > Cancel

Create the Zone File



The screenshot shows the 'New Zone Wizard' window at the 'Zone File' step. The title bar says 'New Zone Wizard' with a close button. Below the title, the section is 'Zone File' with a server icon. The text says 'You can create a new zone file or use a file copied from another DNS server.' The main area asks 'Do you want to create a new zone file or use an existing file that you have copied from another DNS server?'. There are two radio buttons: 'Create a new file with this file name:' (selected) and 'Use this existing file:'. The first option has a text box containing 'tfsi.com.dns'. The second option has an empty text box. Below the text boxes, a note says 'To use this existing file, ensure that it has been copied to the folder %SystemRoot%\system32\dns on this server, and then click Next.' At the bottom, there are three buttons: '< Back', 'Next >' (highlighted with a blue border), and 'Cancel'.

New Zone Wizard

Zone File

You can create a new zone file or use a file copied from another DNS server.

Do you want to create a new zone file or use an existing file that you have copied from another DNS server?

☒ Create a new file with this file name:

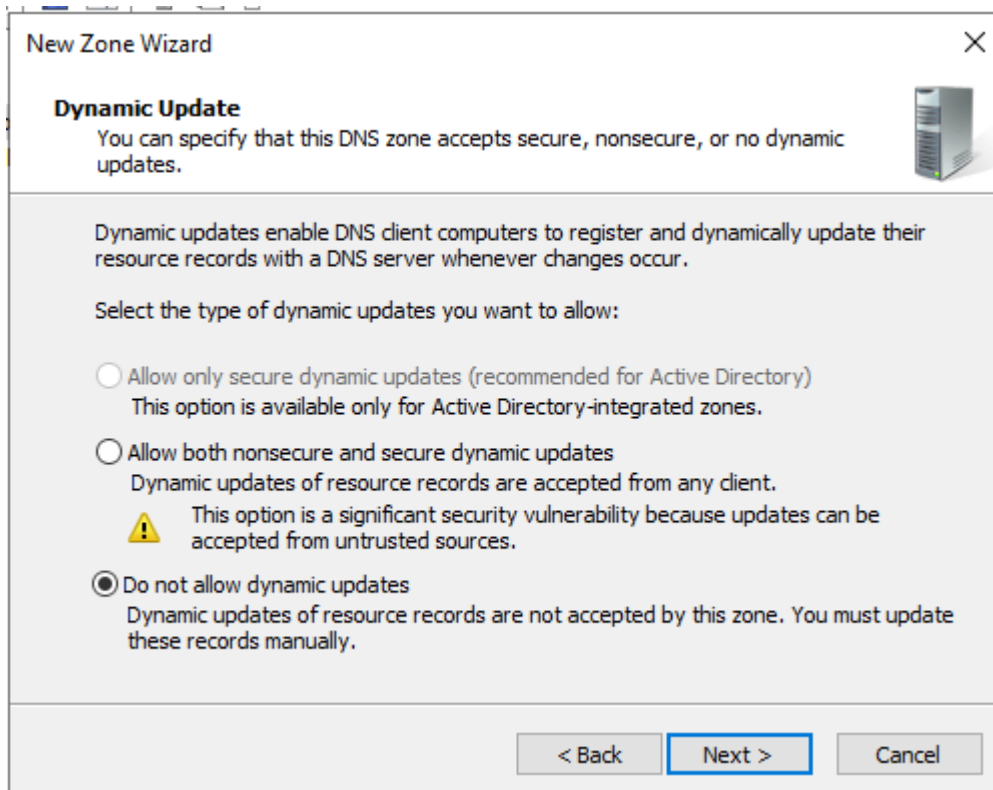
tfsi.com.dns

☐ Use this existing file:

To use this existing file, ensure that it has been copied to the folder %SystemRoot%\system32\dns on this server, and then click Next.

< Back Next > Cancel

Select "Don't allow dynamic updates"



The screenshot shows the 'New Zone Wizard' window at the 'Dynamic Update' step. The title bar says 'New Zone Wizard' with a close button. Below the title, the section is 'Dynamic Update' with a server icon. The text says 'You can specify that this DNS zone accepts secure, nonsecure, or no dynamic updates.' The main area explains 'Dynamic updates enable DNS client computers to register and dynamically update their resource records with a DNS server whenever changes occur.' It then asks 'Select the type of dynamic updates you want to allow:'. There are three radio buttons: 'Allow only secure dynamic updates (recommended for Active Directory)' (with a note 'This option is available only for Active Directory-integrated zones.'), 'Allow both nonsecure and secure dynamic updates' (with a note 'Dynamic updates of resource records are accepted from any client.' and a warning icon 'This option is a significant security vulnerability because updates can be accepted from untrusted sources.'), and 'Do not allow dynamic updates' (selected, with a note 'Dynamic updates of resource records are not accepted by this zone. You must update these records manually.'). At the bottom, there are three buttons: '< Back', 'Next >' (highlighted with a blue border), and 'Cancel'.

New Zone Wizard

Dynamic Update

You can specify that this DNS zone accepts secure, nonsecure, or no dynamic updates.

Dynamic updates enable DNS client computers to register and dynamically update their resource records with a DNS server whenever changes occur.

Select the type of dynamic updates you want to allow:

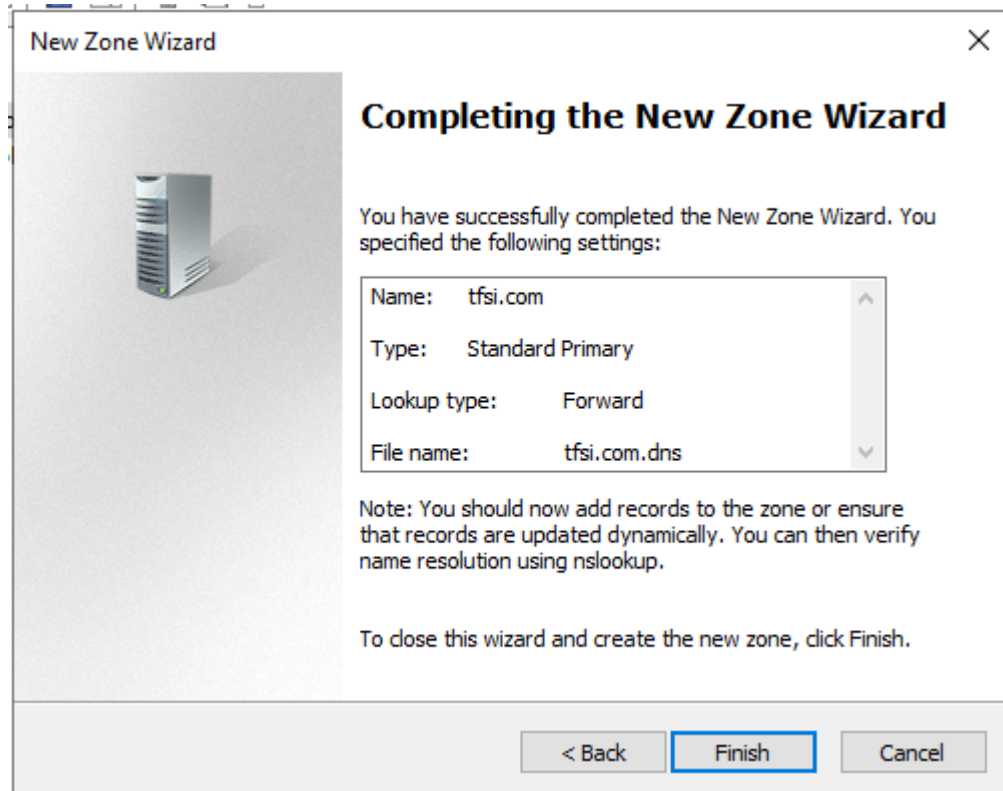
☐ Allow only secure dynamic updates (recommended for Active Directory)
This option is available only for Active Directory-integrated zones.

☐ Allow both nonsecure and secure dynamic updates
Dynamic updates of resource records are accepted from any client.
 This option is a significant security vulnerability because updates can be accepted from untrusted sources.

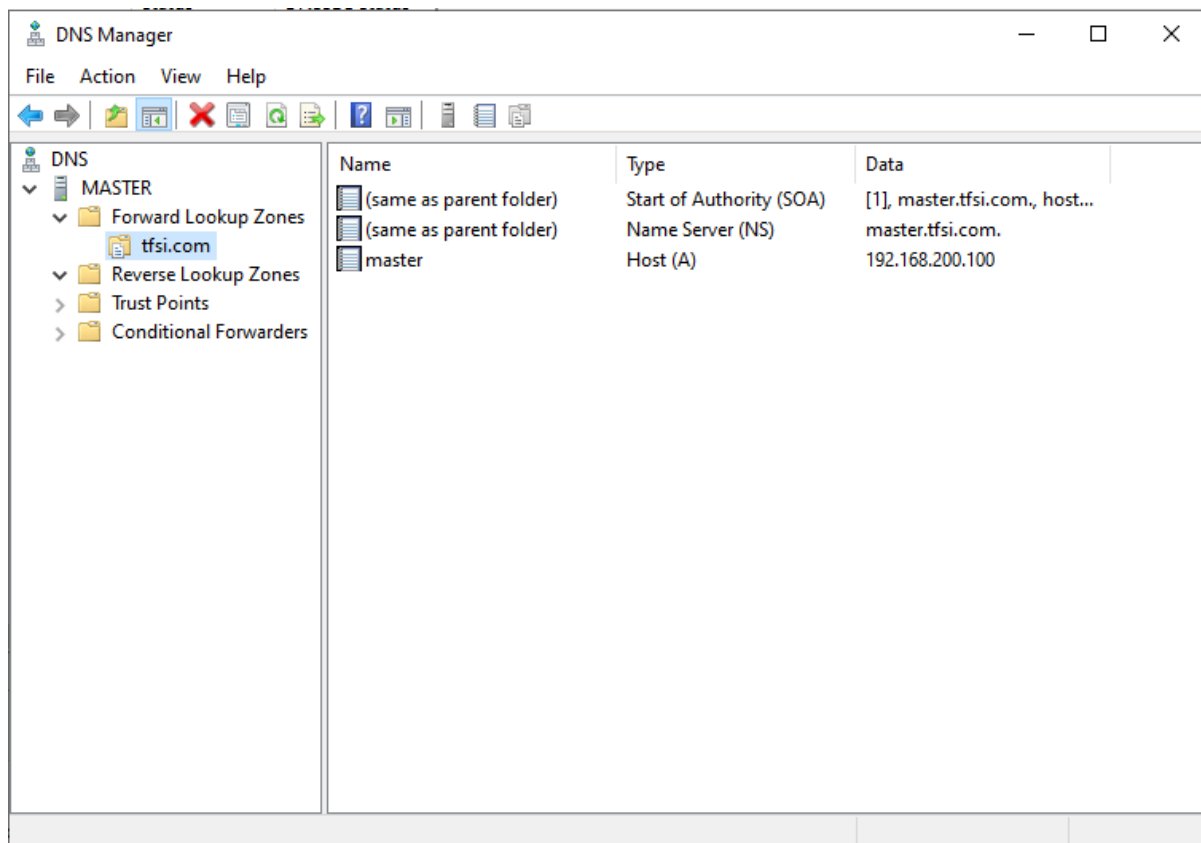
☒ Do not allow dynamic updates
Dynamic updates of resource records are not accepted by this zone. You must update these records manually.

< Back Next > Cancel

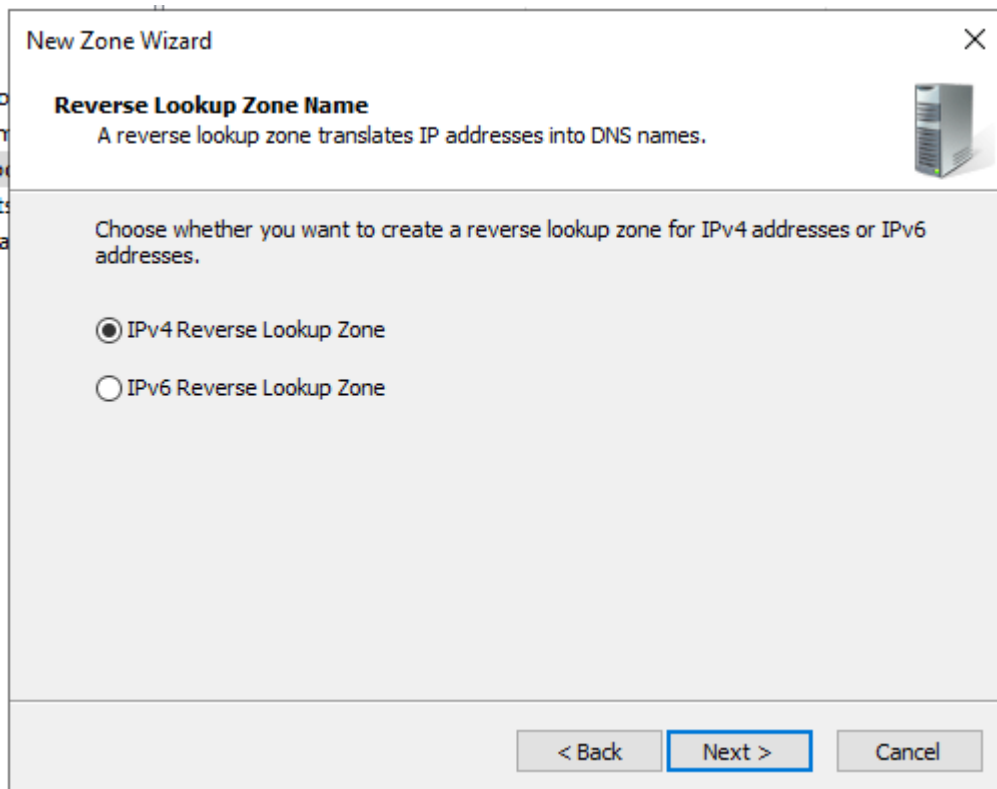
The click on finish



We can see all the records in the DNS Manager

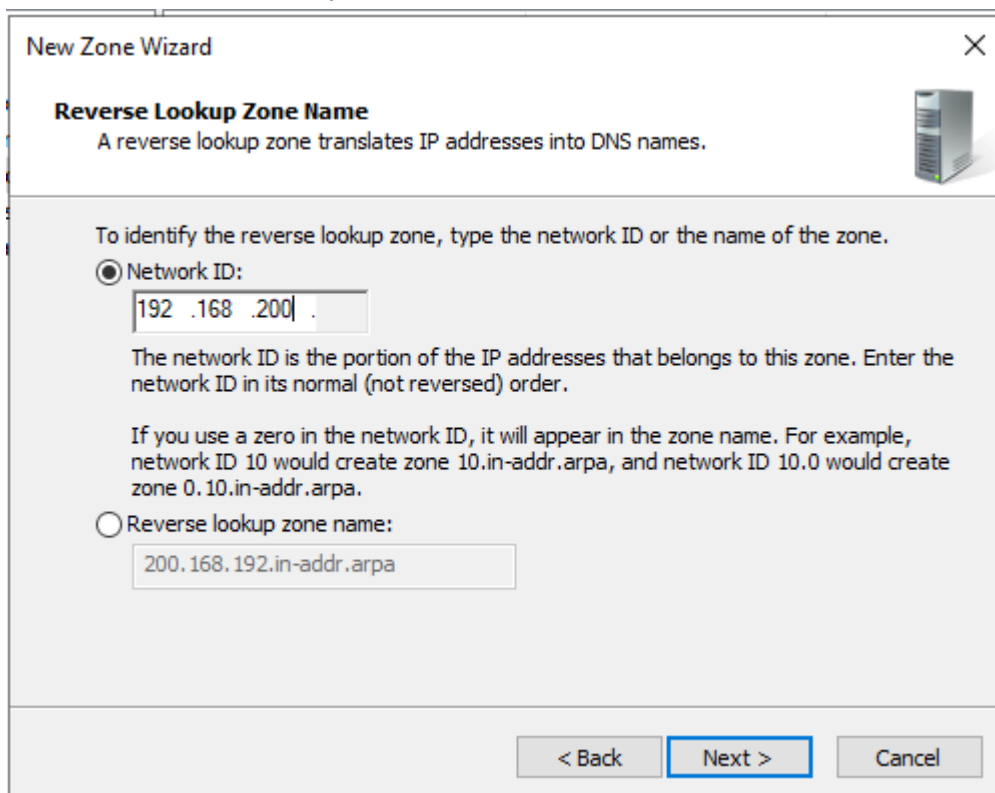


Following the same wizard for new zone, create a new reverse zone
Select IPv4



The screenshot shows the 'New Zone Wizard' window with the title 'New Zone Wizard'. Below the title bar, there is a section titled 'Reverse Lookup Zone Name' with a server icon. The text below the title says 'A reverse lookup zone translates IP addresses into DNS names.' Below this, there is a prompt: 'Choose whether you want to create a reverse lookup zone for IPv4 addresses or IPv6 addresses.' There are two radio buttons: 'IPv4 Reverse Lookup Zone' (which is selected) and 'IPv6 Reverse Lookup Zone'. At the bottom right, there are three buttons: '< Back', 'Next >' (highlighted with a blue border), and 'Cancel'.

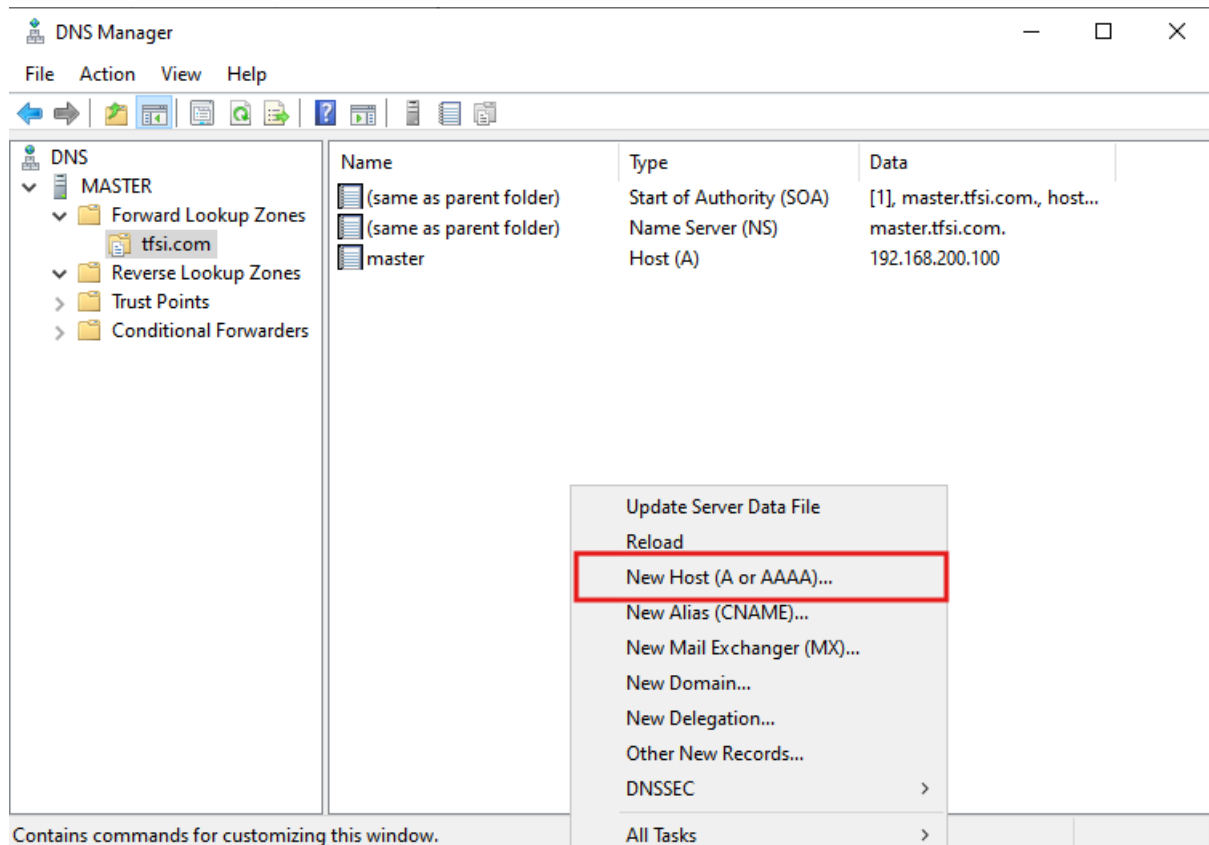
Enter the first 3 octets of your IP



The screenshot shows the 'New Zone Wizard' window with the title 'New Zone Wizard'. Below the title bar, there is a section titled 'Reverse Lookup Zone Name' with a server icon. The text below the title says 'A reverse lookup zone translates IP addresses into DNS names.' Below this, there is a prompt: 'To identify the reverse lookup zone, type the network ID or the name of the zone.' There are two radio buttons: 'Network ID:' (which is selected) and 'Reverse lookup zone name:'. The 'Network ID:' radio button has a text box next to it containing '192 .168 .200'. Below this text box, there is a paragraph of text: 'The network ID is the portion of the IP addresses that belongs to this zone. Enter the network ID in its normal (not reversed) order.' Below this paragraph, there is another paragraph: 'If you use a zero in the network ID, it will appear in the zone name. For example, network ID 10 would create zone 10.in-addr.arpa, and network ID 10.0 would create zone 0.10.in-addr.arpa.' The 'Reverse lookup zone name:' radio button has a text box next to it containing '200.168.192.in-addr.arpa'. At the bottom right, there are three buttons: '< Back', 'Next >' (highlighted with a blue border), and 'Cancel'.

Now click on next as proceed as usual & Finish

To Create a new record, in the Forward/Reverse Zone window
Right click and select “New Host” option



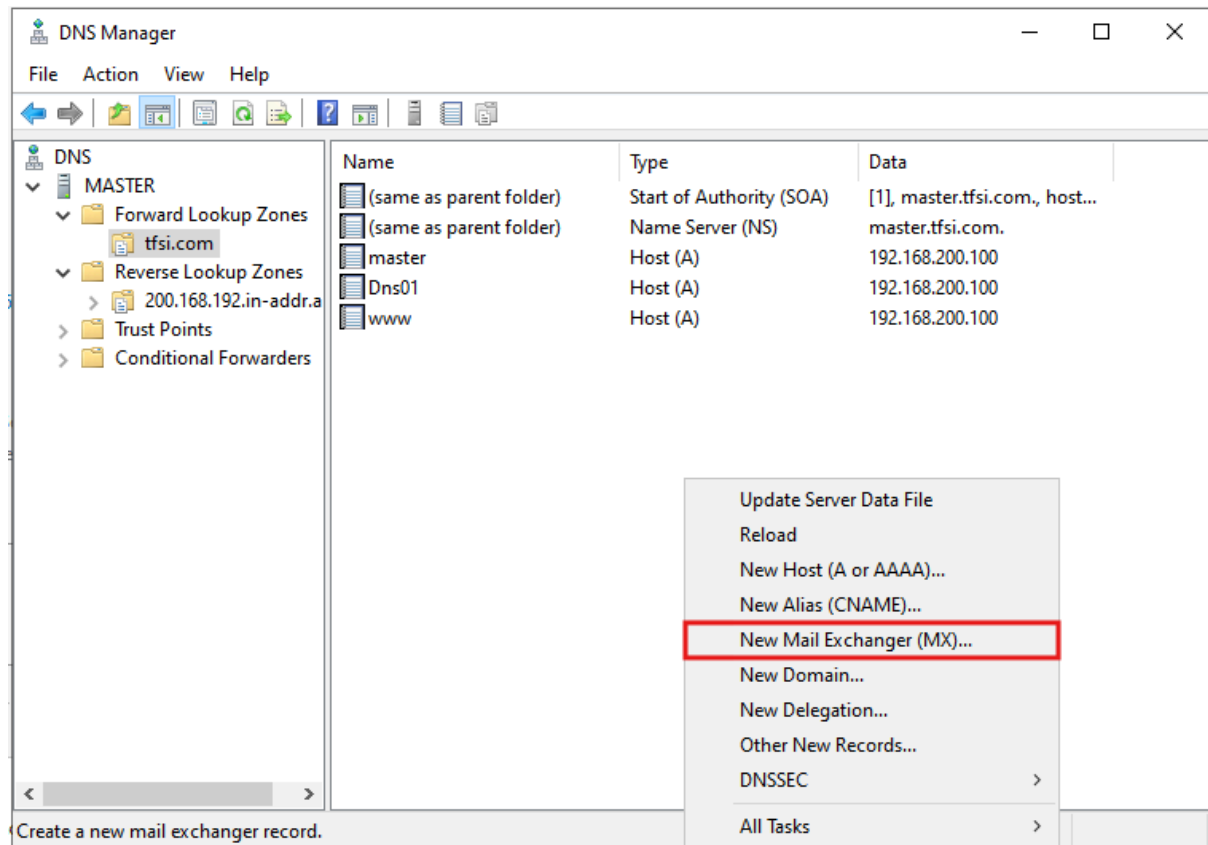
Enter the details

The 'New Host' dialog box is shown with the following fields and options:

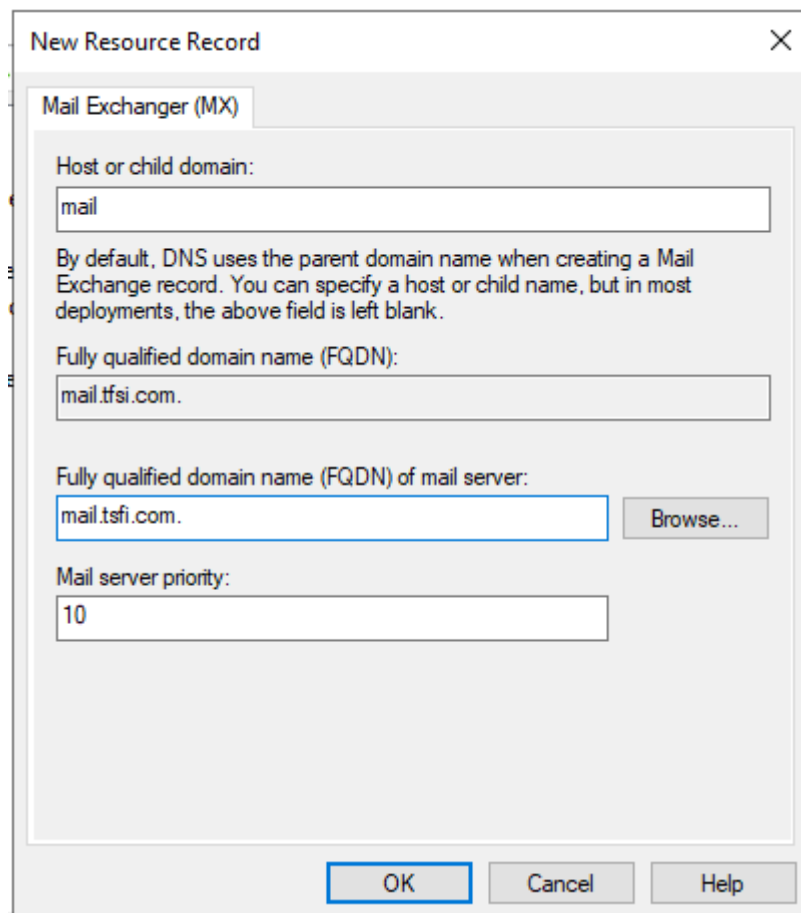
- Name (uses parent domain name if blank):** Dns01
- Fully qualified domain name (FQDN):** Dns01.tfsi.com.
- IP address:** 192.168.200.100
- ☒ Create associated pointer (PTR) record

Buttons at the bottom: Add Host, Cancel

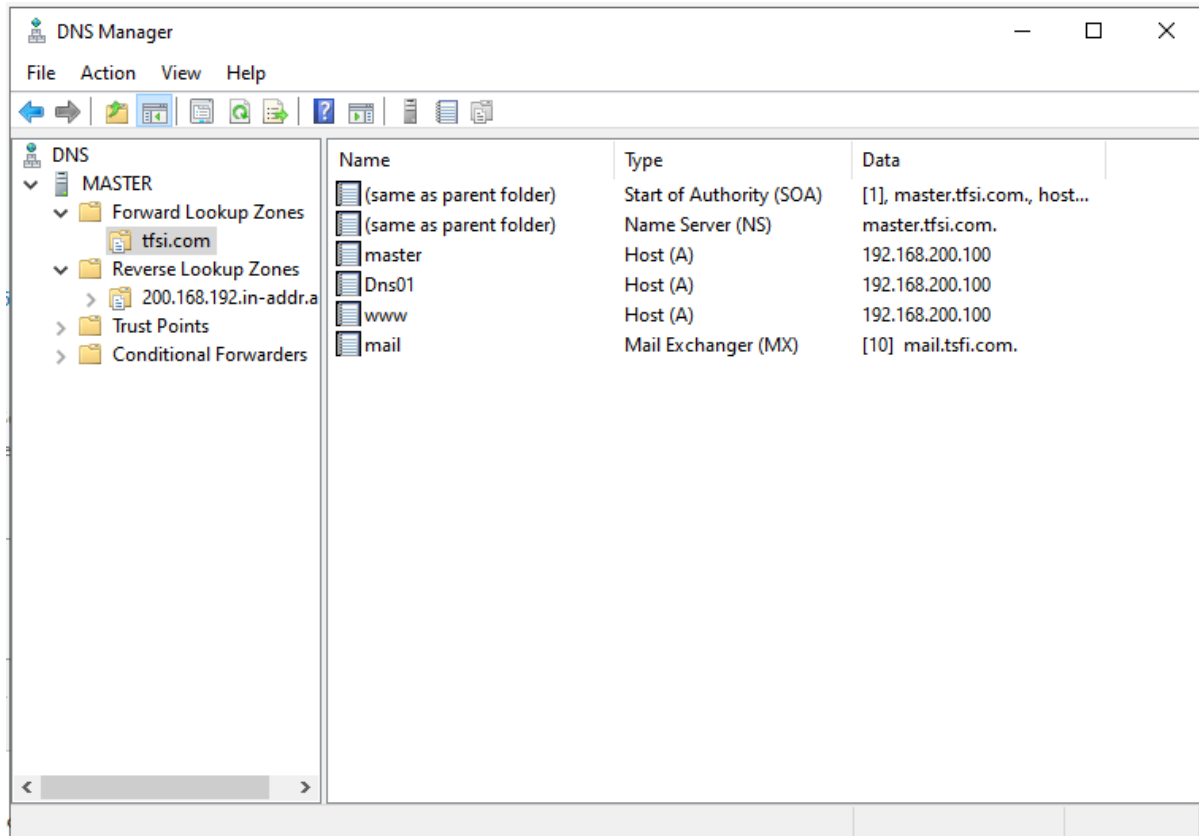
For the mail record, use the new MX Record option from the right click context menu



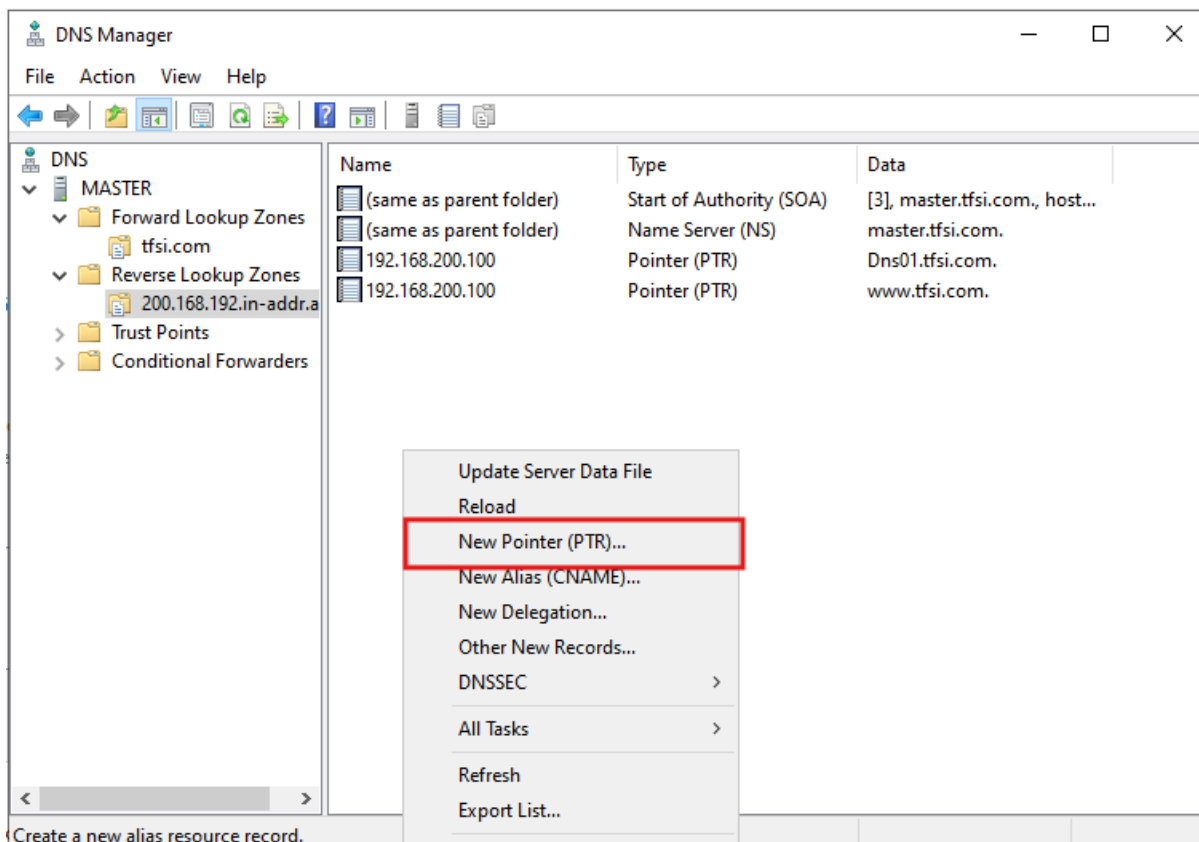
Enter these details



All forward zone records were created successfully



For the PTR Record in Reverse Zone, select the “New PTR Record” from right click context menu.



Fill these details

New Resource Record

Pointer (PTR)

Host IP Address:

192.168.200.100

Fully qualified domain name (FQDN):

100.200.168.192.in-addr.arpa

Host name:

mail.tfsi.com

Browse...

OK

Cancel

All records in the reverse zone

DNS Manager

FileActionViewHelp

DNS

MASTER

Forward Lookup Zones

tfsi.com

Reverse Lookup Zones

200.168.192.in-addr.a

Trust Points

Conditional Forwarders

Name	Type	Data
(same as parent folder)	Start of Authority (SOA)	[3], master.tfsi.com., host...
(same as parent folder)	Name Server (NS)	master.tfsi.com.
192.168.200.100	Pointer (PTR)	Dns01.tfsi.com.
192.168.200.100	Pointer (PTR)	www.tfsi.com.
192.168.200.100	Pointer (PTR)	mail.tfsi.com

Verified the DNS is working

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.20348.587]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>nslookup www.tfsi.com
Server:    www.tfsi.com
Address:   192.168.200.100

Name:      www.tfsi.com
Address:   192.168.200.100

C:\Users\Administrator>nslookup Dns01.tfsi.com
Server:    mail.tfsi.com
Address:   192.168.200.100

Name:      Dns01.tfsi.com
Address:   192.168.200.100

C:\Users\Administrator>

C:\Users\Administrator>nslookup -type=a 192.168.200.100
Server:    Dns01.tfsi.com
Address:   192.168.200.100

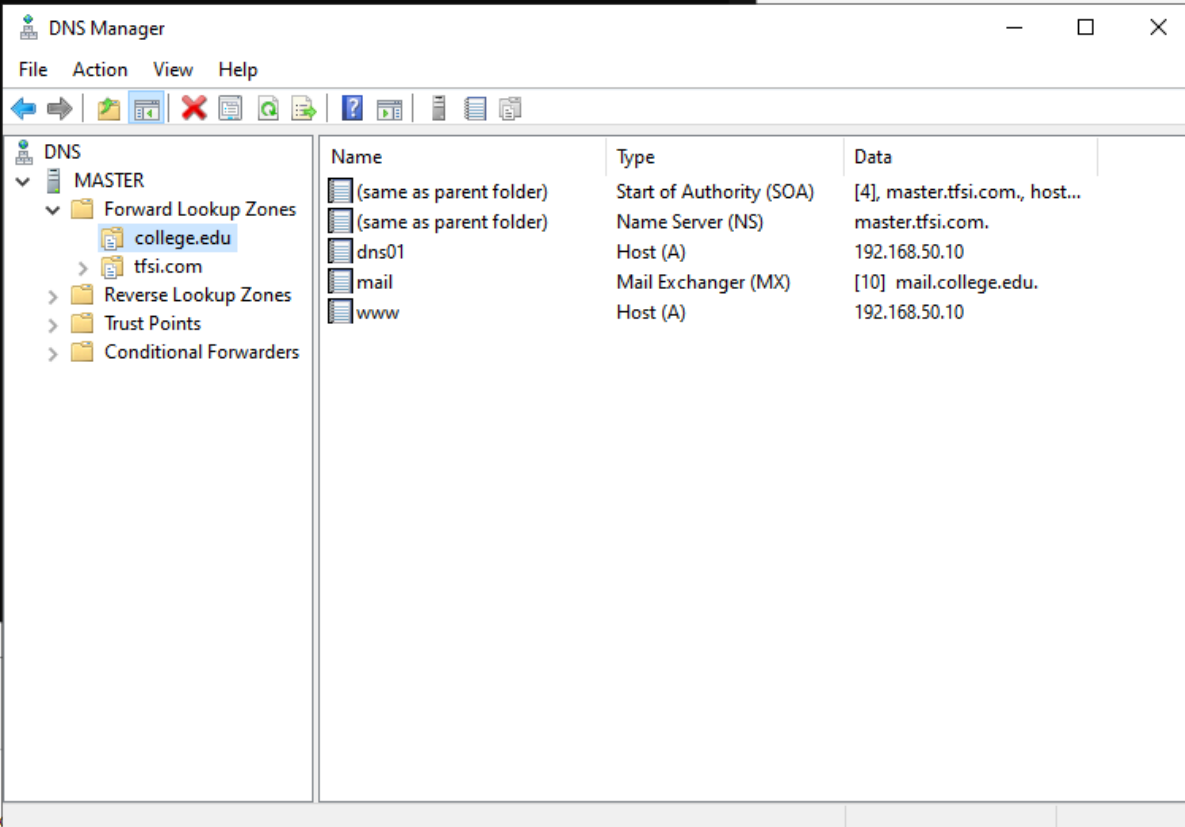
Name:      www.tfsi.com
Address:   192.168.200.100

C:\Users\Administrator>nslookup -type=mx 192.168.200.100
Server:    mail.tfsi.com
Address:   192.168.200.100

DNS request timed out.
    timeout was 2 seconds.
*** Request to mail.tfsi.com timed-out

C:\Users\Administrator>
```

Question 2

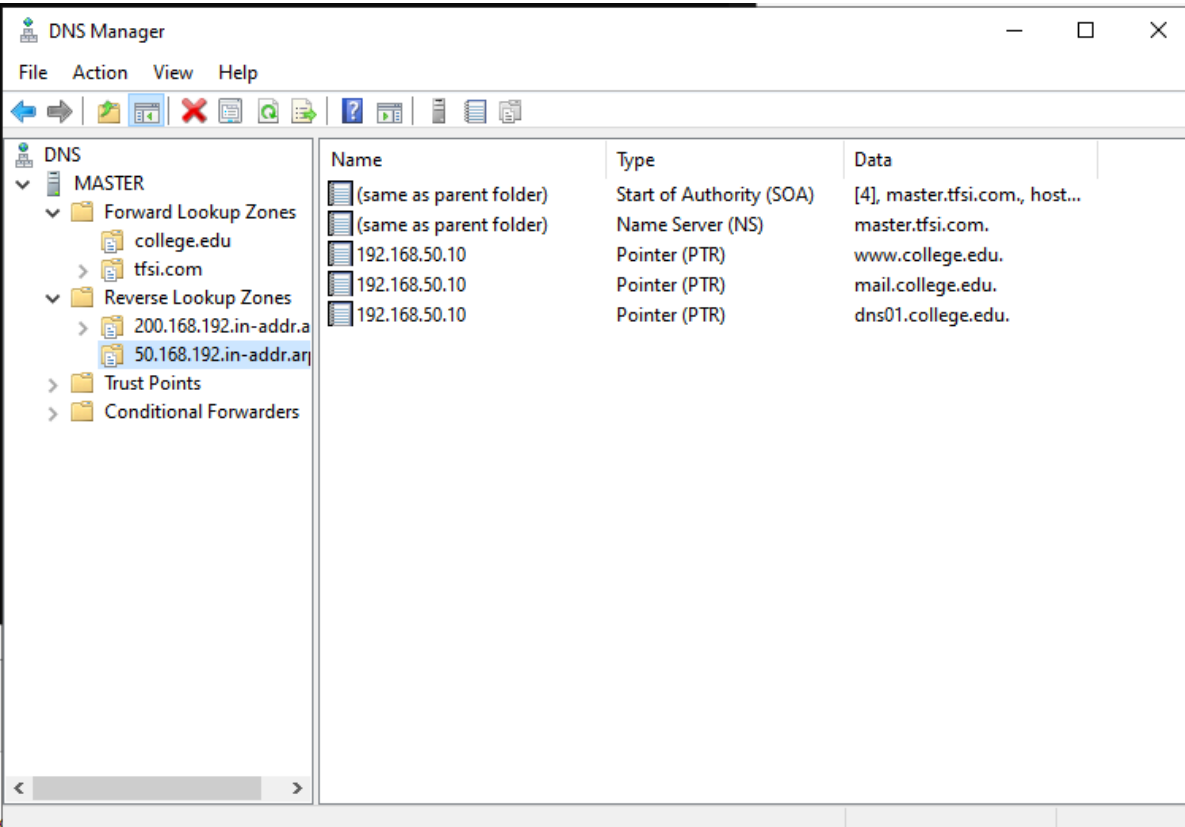


The screenshot shows the DNS Manager console tree with the following structure:

- DNS
 - MASTER
 - Forward Lookup Zones
 - college.edu (selected)
 - tfsi.com
 - Reverse Lookup Zones
 - Trust Points
 - Conditional Forwarders

The main pane displays the following records for the selected zone:

Name	Type	Data
(same as parent folder)	Start of Authority (SOA)	[4], master.tfsi.com., host...
(same as parent folder)	Name Server (NS)	master.tfsi.com.
dns01	Host (A)	192.168.50.10
mail	Mail Exchanger (MX)	[10] mail.college.edu.
www	Host (A)	192.168.50.10



The screenshot shows the DNS Manager console tree with the following structure:

- DNS
 - MASTER
 - Forward Lookup Zones
 - college.edu
 - tfsi.com
 - Reverse Lookup Zones
 - 200.168.192.in-addr.a
 - 50.168.192.in-addr.arpa (selected)
 - Trust Points
 - Conditional Forwarders

The main pane displays the following records for the selected zone:

Name	Type	Data
(same as parent folder)	Start of Authority (SOA)	[4], master.tfsi.com., host...
(same as parent folder)	Name Server (NS)	master.tfsi.com.
192.168.50.10	Pointer (PTR)	www.college.edu.
192.168.50.10	Pointer (PTR)	mail.college.edu.
192.168.50.10	Pointer (PTR)	dns01.college.edu.

```
C:\Users\Administrator>nslookup -type=a www.college.edu
Server: www.tfsi.com
Address: 192.168.200.100

Name: www.college.edu
Address: 192.168.50.10

C:\Users\Administrator>
```

```
C:\Users\Administrator>nslookup -type=mx mail.college.edu
Server: Dns01.tfsi.com
Address: 192.168.200.100

mail.college.edu      MX preference = 10, mail exchanger = mail.college.edu

C:\Users\Administrator>
```

```
C:\Users\Administrator>nslookup -type=ptr 192.168.50.10
Server: www.tfsi.com
Address: 192.168.200.100

10.50.168.192.in-addr.arpa    name = www.college.edu
10.50.168.192.in-addr.arpa    name = mail.college.edu
10.50.168.192.in-addr.arpa    name = dns01.college.edu

C:\Users\Administrator>
```